WEEK 2 TASK REPORT

Hands-on Practice in Mechanical Design: Modelling, Tolerancing and Documentation of a 90° Flanged Pipe Elbow

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Task	Week 2: Hands-on Practice in Mechanical Design (SolidWorks, GD&T, design)
Date	29 October 2026
Software	SolidWorks Design Standard 2026 SP4.1 (Academic)
Part and drawing	Part id0.6; drawing “model01 drafting”, A3, scale 1:5

1. Objective and Summary

The task asks for a guided, real-world style exercise using SolidWorks, GD&T and design, with the steps documented. For this exercise I took an existing SolidWorks part, a 90° flanged pipe elbow (part id0.6), and developed it from a dimensioned model into a manufacturing-ready package: material selection, a design-for-manufacturing (DFM) review, GD&T and size tolerancing, supporting calculations and a documented A3 drawing.

Every decision in this report is backed by a number or a worked calculation, and each section states why the choice was made, so the approach can be followed and checked.

Table 1. Summary of the exercise

Item	Result
Part	90° flanged pipe elbow; two square flanges 360 × 360 × 50 mm; 4 × Ø30 holes per flange; bore Ø210; OD Ø290; bend R260
Material	AISI 1020 plain carbon steel (assigned in SolidWorks)
Calculated mass	≈ 247 kg (hand calculation, Section 4.3)
Wall thickness	40 mm in the model; minimum needed at 1 MPa with a safety factor of 3 is 0.9 mm (Section 5.2)
Improvement proposed	10 mm wall: ≈ 111 kg (−55%), hoop stress 11.0 MPa (Section 5.3)
GD&T	Datums A to D; flatness 0.1; perpendicularity 0.2; position Ø1.0 at MMC; bore Ø210 H8; holes Ø30 H12
Assembly margin	2.0 mm at worst case (3.0 mm allowed, 1.0 mm used; Section 7.4)
Deliverables	This report and the A3 drawing (PNG and PDF)

2. Design Brief, Geometry and Assumptions

The elbow turns a flanged pipe run through 90°. Each end has a square flange with four bolt holes for a gasketed bolted joint. The geometry below is taken from the SolidWorks model and its dimensioned drawing.

Table 2. Geometry from the SolidWorks model

Parameter	Value	Note
Flange square side	360 mm	Both flanges
Flange thickness	50 mm	Both flanges
Bolt hole spacing (square pattern)	260 mm	4 holes per flange
Bolt hole diameter	Ø30 mm	Through
Spot-face diameter	Ø60 mm	Around each hole
Bore diameter	Ø210 mm	Through both flanges and the body
Pipe outside diameter	Ø290 mm	
Bend radius	R260 mm	90° bend
Straight length	150 mm	Each side of the bend
Corner radius / fillets	R20 / R10	Flange corners / blends

Where the model does not state a value, the following assumptions are used. They are listed so that every calculation can be traced and re-run if a value changes.

Table 3. Assumptions

ID	Assumption	Basis and effect
A1	Design pressure 1 MPa	Assumed for the wall check; replace with the real duty if it differs
A2	M27 bolts in Ø30 clearance holes	Ø30 is the ISO 273 medium-series clearance hole for M27; used in the floating-fastener calculation
A3	AISI 1020: density 7,870 kg/m³, yield ≈ 350 MPa, tensile ≈ 420 MPa	Typical published as-rolled values; confirm against the material certificate or the SolidWorks library
A4	Safety factor 3	Design choice for the check
A5	150 mm straights run from each flange's inner face to the bend; R260 is the centreline radius	Affects the mass estimate only

3. Step-by-Step Modelling Process in SolidWorks

Model id0.6 was built in SolidWorks Design Standard 2026 SP4.1 (Academic). The feature tree, in build order, is: Sweep1, Boss-Extrude1, Cut-Extrude1, Boss-Extrude3, Cut-Extrude3, Cut-Extrude4, Fillet2, Fillet3, Cut-Extrude5 and Chamfer2, with the material AISI 1020 assigned. The table groups these into the working steps and gives the reason behind each.

Table 4. Modelling steps

Step	Feature(s)	What it creates	Why this way (best practice)
1	Material: AISI 1020	Assigns density and strength data to the part	Set first so mass properties and any later analysis use the right material
2	Sweep1	The curved pipe body along the 90° path (R260)	One sweep gives a constant section and a smooth bend; the path drives the shape

Step	Feature(s)	What it creates	Why this way (best practice)
3	Boss-Extrude1, Boss-Extrude3	The two 360 × 360 × 50 mm square flanges	One extrude per flange, so each thickness can be edited on its own
4	Cut-Extrude1, 3, 4, 5	Ø210 bore, 4 × Ø30 bolt holes per flange, Ø60 spot faces	Cuts follow the solid features, so they remove material from the finished shape
5	Fillet2, Fillet3	R10 blends and R20 flange corners	Radii are added late to keep sketches simple and avoid rebuild errors
6	Chamfer2	Edge break	Removes a sharp edge, as required by the “break sharp edges” note

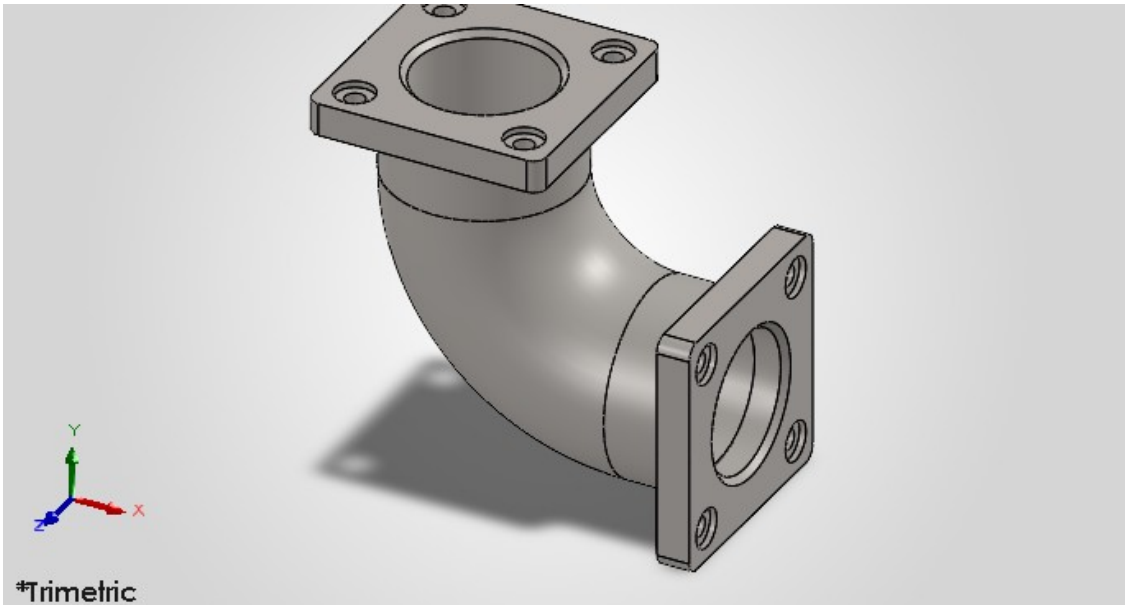


Figure 1. Trimetric view of the finished elbow (part id0.6) in SolidWorks, showing the two square flanges, the bolt holes with spot faces, and the 90° bend.

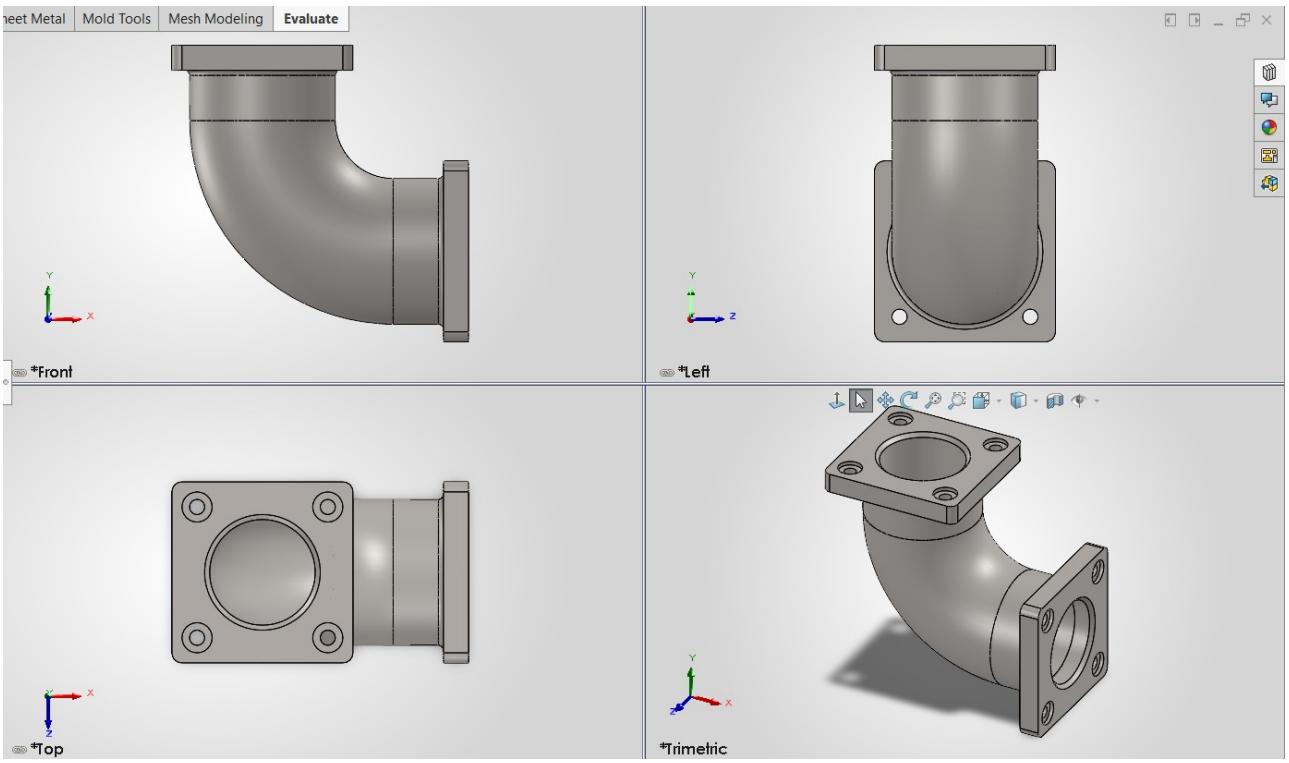


Figure 2. Working setup: the feature tree (left) and the standard views of id0.6 in SolidWorks Design Standard 2026 SP4.1. AISI 1020 appears in the tree as the assigned material.

4. Material Selection

4.1 Candidate materials

Four candidates were compared for a flanged elbow that must be machined on its flange faces, bolted, and possibly welded. The strength and density figures are typical published values; the mass column applies the density ratio to the AISI 1020 estimate for the same geometry.

Table 5. Material comparison

Property	AISI 1020 (as rolled)	Grey cast iron (A48 Class 30)	Stainless 304 (annealed)	Aluminium 6061-T6
Density (kg/m³)	7,870	≈ 7,150	≈ 8,000	≈ 2,700
Yield strength (MPa)	≈ 350	No distinct yield	≥ 205	≈ 276
Tensile strength (MPa)	≈ 420	≥ 207	≥ 515	≈ 310
Machinability	Good	Good, brittle chips	Fair (work hardens)	Very good
Weldability	Excellent	Poor	Good	Fair, needs care
Corrosion resistance	Low, needs a coating	Moderate	High	High
Relative cost	Low	Low	High	Medium
Mass for this geometry	≈ 247 kg	≈ 225 kg	≈ 251 kg	≈ 85 kg

4.2 Weighted decision

Each material was scored from 1 (poor) to 5 (excellent) against five criteria. The weights reflect this part: it is machined and bolted, so cost and manufacturability matter most. The scores are design judgements based on the comparison in Section 4.1.

Table 6. Weighted scoring (maximum 5.00)

Material	Strength (25%)	Manufact. (25%)	Corrosion (15%)	Cost (25%)	Weight (10%)	Score
AISI 1020	4	5	2	5	3	4.10
Grey cast iron	2	3	3	5	3	3.25
Stainless 304	3	3	5	2	3	3.05
Aluminium 6061-T6	3	4	4	3	5	3.60

Decision: AISI 1020. It scores highest (4.10) and was already assigned in the model. It is ductile, weldable (so a welded-flange route stays open), easy to machine on the flange faces and low in cost. Its yield strength of about 350 MPa is far above the stress in this part (Section 5.2). Its weakness is corrosion, so the part needs paint or zinc coating, and stainless 304 should be chosen instead if the fluid is corrosive.

4.3 Mass estimate (worked example)

The mass is found by adding the volume of the pipe body to the volume of the two flanges, using the assumptions in Table 3.

Table 7. Mass calculation for AISI 1020

Quantity	Calculation	Result
Pipe cross-section area	$\pi/4 \times (290^2 - 210^2)$	31,416 mm²
Centreline length of the body	150 + (π/2 × 260) + 150	708.4 mm
Pipe body volume	31,416 × 708.4	22,255,264 mm³
One flange volume	$[360^2 - 4(1 - \pi/4)\cdot 20^2 - \pi/4\cdot 210^2] \times 50 - 4 \times \pi/4\cdot 30^2 \times 50$	4,589,657 mm³
Total volume	22,255,264 + 2 × 4,589,657	31,434,578 mm³ = 0.03143 m³
Mass	0.03143 × 7,870	**≈ 247.4 kg**

The estimate leaves out the R10 blends, the chamfer and the spot-face depth. These change the result by well under 1% unless the spot faces are very deep. A mass of this size needs crane handling, so lifting provisions should be planned.

5. Engineering Calculations and Design Analysis

5.1 Derived geometry

Table 8. Derived geometry

Quantity	Formula	Value
Wall thickness	(290 − 210) / 2	40 mm
Bolt circle diameter	260 × √2	367.7 mm

Quantity	Formula	Value
Hole centre to bore axis	Bolt circle / 2	183.85 mm
Edge distance (hole centre to flange edge)	(360 − 260) / 2	50 mm
Material between hole edge and bore	183.85 − 105 − 15	63.85 mm
Gap from hole edge to pipe OD	183.85 − 145 − 15	23.85 mm
Gap from spot-face edge to pipe OD	183.85 − 145 − 30	8.85 mm
Spot-face clearance round an M27 washer (OD 50)	(60 − 50) / 2	5 mm

The 50 mm edge distance is more than three times the hole radius, which leaves ample material at the flange edge. The 8.85 mm gap between the spot face and the pipe OD is tight and is taken up again in Section 6.

5.2 Wall thickness check under internal pressure

The allowable stress is the yield strength divided by the safety factor: 350 / 3 = 116.7 MPa. For a thin wall, the minimum thickness at 1 MPa is $t = P \cdot D / (2\sigma) = 1 \times 210 / (2 \times 116.7)$ = **0.90 mm**. The model's 40 mm wall is therefore about 44 times thicker than pressure alone requires.

Because 40 mm is not a thin wall, the thick-wall (Lamé) hoop stress at the bore was also checked: $\sigma = P (r_o^2 + r_i^2) / (r_o^2 - r_i^2) = 1 \times (145^2 + 105^2) / (145^2 - 105^2)$ = **3.21 MPa**. Against a yield of 350 MPa this is a factor of 109. The wall is set by handling and manufacturing, not by pressure.

5.3 Improvement proposal: thinner wall

Keeping the Ø210 bore and flanges unchanged, the pipe OD is reduced from Ø290 to Ø230, giving a 10 mm wall. This is still more than eleven times the 0.90 mm pressure minimum.

Table 9. Effect of reducing the wall to 10 mm

Quantity	Current (40 mm wall)	Proposed (10 mm wall)	Change
Pipe OD	Ø290	Ø230	−60 mm
Pipe cross-section area	31,416 mm²	6,912 mm²	−78%
Calculated mass	247.4 kg	110.8 kg	−136.6 kg (−55.2%)
Lamé hoop stress at 1 MPa	3.21 MPa	11.02 MPa	Still low
Safety factor on yield (350 MPa)	109.2	31.8	Still above 30
Gap, spot-face edge to pipe OD	8.85 mm	38.85 mm	+30 mm

The thinner wall removes more than half the mass and also opens the tool clearance at the bolt heads from 8.85 mm to 38.85 mm. Before adopting it, the bend and flange-to-pipe junction should be checked by finite element analysis under pressure and bolt preload, and a corrosion allowance and the casting or welding minimum thickness should be agreed with the manufacturer. These checks are outside the scope of this exercise.

6. Design for Manufacturability (DFM) Review

The elbow can be made by sand casting followed by machining, or by fabrication (a standard bend with welded flanges). The review below lists what each observation means for cost and quality, and the action taken or recommended.

Table 10. DFM observations and actions

Observation	Why it matters	Action
40 mm wall and 50 mm flange (ratio 1.25) meet at the junction	Thick sections cool slowly in a casting and can leave shrinkage porosity; they also need heavy weld preparation	Reduce the wall (Section 5.3); keep the R10 blend at the junction
Spot-face edge only 8.85 mm from the pipe OD	Limits access for the spot-facing cutter and for the socket or spanner on the bolt head	A 10 mm wall raises the gap to 38.85 mm
Flange faces need Ra 3.2 and flatness 0.1	Cast or flame-cut faces cannot reach this	Leave a machining allowance on both faces (value agreed with the foundry)
Perpendicularity 0.2 between the two flange faces	Hard to hold if each face is machined in a separate set-up	Machine both faces in one fixture where possible
Holes toleranced Ø30 H12	IT12 is achievable by drilling, so no reaming is needed	Use plain drilling; control location with the position tolerance instead
Bores toleranced Ø210 H8	IT8 needs boring or reaming, which adds cost	Apply H8 only to the bores that act as datums (B and D)
No draft angle shown on the drawing	Cast surfaces need draft so the pattern can be withdrawn	If cast, add draft (typically 1° to 3°) on unmachined surfaces
Mass of about 247 kg	Cannot be handled by hand	Plan crane lifting; a lighter wall also helps

Table 11. Manufacturing route comparison

Route	Strengths	Weaknesses
Sand casting + machining	Suits medium to large batches; one-piece part with a smooth bend	Pattern cost; thick sections risk porosity; machining allowance needed
Fabrication (bend + welded flanges)	No pattern; AISI 1020 welds well; good for one-off parts	Welding distortion can spoil flange flatness and perpendicularity; more inspection

7. GD&T and Tolerancing

7.1 Datum scheme

The elbow has two mating interfaces at 90° to each other. Each flange must match its own mating flange, so each gets its own datum frame, and the perpendicularity tolerance ties the two frames together.

Table 12. Datum reference frame

Datum	Feature	Reason
A	Flange 1 mating face	Primary datum: the face that seals against the gasket and mating flange; it fixes the part in the mounting plane

Datum	Feature	Reason
B	Ø210 bore axis, flange 1	Secondary datum: centres the part on the pipe axis and locates the hole pattern with A
C	Flange 2 mating face	Primary datum for the second flange; its orientation to A is controlled by perpendicularity
D	Ø210 bore axis, flange 2	Locates the flange 2 hole pattern with C

7.2 Callouts and their purpose

Table 13. GD&T callouts on the drawing

Feature	Callout	Value	Purpose
Flange 1 face (A)	Flatness	0.1 mm	Gasket seal; datum quality
Flange 2 face (C)	Flatness; perpendicularity to A	0.1 mm; 0.2 mm	Seal; accuracy of the 90° turn
Bores Ø210 (B, D)	Size H8	210.000 to 210.072	Datum axes; centring
4 × Ø30 holes, flange 1	Position at MMC to A | B	Ø1.0 (M)	Bolts line up with the mating flange
4 × Ø30 holes, flange 2	Position at MMC to C | D	Ø1.0 (M)	As above, for flange 2
Hole size	H12	30.000 to 30.210	Drilled hole
Ø60 spot face	Size	±0.5 mm	Seat for bolt head and washer
Flange faces	Surface finish	Ra 3.2 µm	Gasket seating; Ra 12.5 elsewhere

7.3 Size tolerances and where they come from

Fits come from ISO 286-1 and the general linear tolerances from ISO 2768-m for the relevant size range. The flatness and perpendicularity values are tighter than the general geometrical tolerances of ISO 2768-K, so they are called out explicitly.

Table 14. Size tolerances

Dimension	Tolerance	Basis
Bore Ø210	H8: +0.072 / 0	IT8 = 72 µm for 180 to 250 mm
Hole Ø30	H12: +0.210 / 0	IT12 = 210 µm for 18 to 30 mm
Flange thickness 50	±0.3	ISO 2768-m, 30 to 120 mm
Flange width 360, straight 150, OD Ø290	±0.5	ISO 2768-m, 120 to 400 mm
Bend radius R260	±1.0	ISO 2768-m, radii over 6 mm

7.4 Worked examples

Floating fastener. Both flanges have clearance holes, so the bolt floats. The largest position tolerance that still lets the bolt fit is T = H − F, where H is the smallest hole (30.000 mm) and F is the largest bolt

(27 mm for M27): T = 30 − 27 = **3.0 mm**. The drawing specifies Ø1.0 mm, which is 33% of that limit. This keeps the flanges well aligned without paying for a very tight tolerance.

MMC bonus. With the (M) modifier, the position tolerance grows as the hole gets larger. A hole made at the largest size of Ø30.210 allows Ø1.0 + 0.210 = **Ø1.21 mm**. Parts that would fail a fixed Ø1.0 zone can pass when the hole is oversize, which lowers scrap without hurting the fit.

Assembly margin. If both flanges have holes at the worst-case position (each off by up to Ø1.0), the holes can be offset by at most 1.0 mm. Against the 3.0 mm clearance, that leaves **3.0 − 1.0 = 2.0 mm** of margin.

Bore limits. For Ø210 H8 the lower limit is 210.000 mm and the upper limit is 210.000 + 0.072 = **210.072 mm**.

7.5 Inspection plan

Table 15. How each characteristic would be checked

Characteristic	Method	Note
Flatness 0.1 on flange faces	Surface plate and dial indicator, or CMM	Measure across the whole face
Perpendicularity 0.2 to A	CMM, or square and indicator on a surface plate	Both faces in one setup
Bore Ø210 H8	Bore gauge or go/no-go plug gauge	Check both bores
Hole position Ø1.0 (M)	CMM, or a fixed functional gauge	A functional gauge suits MMC
Surface finish Ra 3.2	Roughness comparator or profilometer	Flange faces only

8. Drawing Documentation

The drawing was developed in two stages. The first stage was a dimensioned SolidWorks drawing of the elbow (Figure 3). It showed the shape but was not ready for manufacture. The second stage added what a manufacturer needs (Figure 4 and the detail views that follow). The final drawing was prepared from the model's dimensions as a documented manufacturing drawing; the tolerances and GD&T on it are my proposals for this exercise. The 3D view of the part is in Figure 1.

Table 16. Stage 1 compared with the final drawing

Aspect	Stage 1 drawing	Final drawing
Title block	Name, size and scale only	Title, drawing number, material, scale, size, mass, finish, tolerance, drawn by, date
Tolerances	None	Size tolerances from ISO 286-1 and ISO 2768-m
GD&T	None	Datums A to D; flatness; perpendicularity; position at MMC
Views	Three orthographic views and an isometric	Section A-A showing the wall; plan view with the cutting-plane line; flange 2 face view
Hole notes	Diameters only	4X Ø30 H12 with position, and the Ø60 spot face
Notes	None	General notes, basic-dimension convention, surface finish, fastener assumption

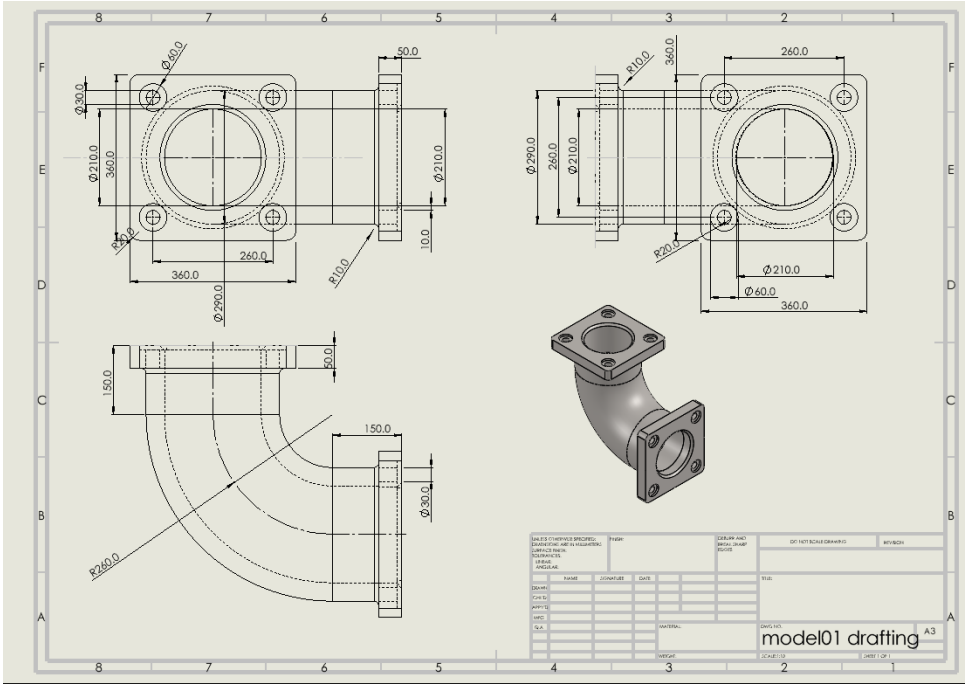


Figure 3. Stage 1: the dimensioned SolidWorks drawing (model01 drafting) before tolerances, GD&T and a complete title block were added.

8.1 Final A3 Manufacturing Drawing

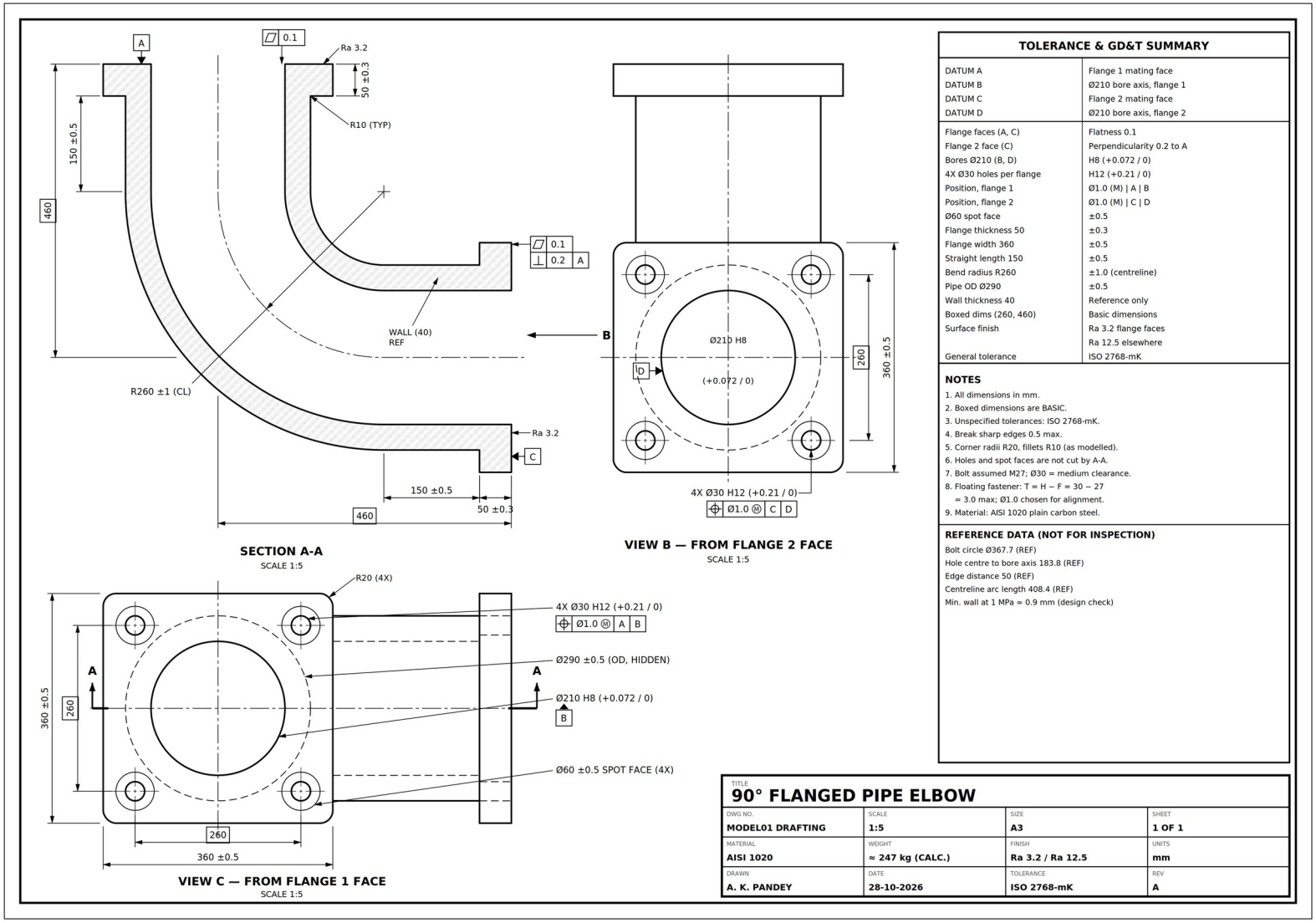


Figure 4. Final drawing: Section A-A, flange 1 face view (C), flange 2 face view (B), tolerance and GD&T summary, notes and title block. A3 sheet, scale 1:5, dimensions in mm.

8.2 Detail views

The views below are enlarged from the final drawing so the dimensions, tolerances and GD&T frames can be read.

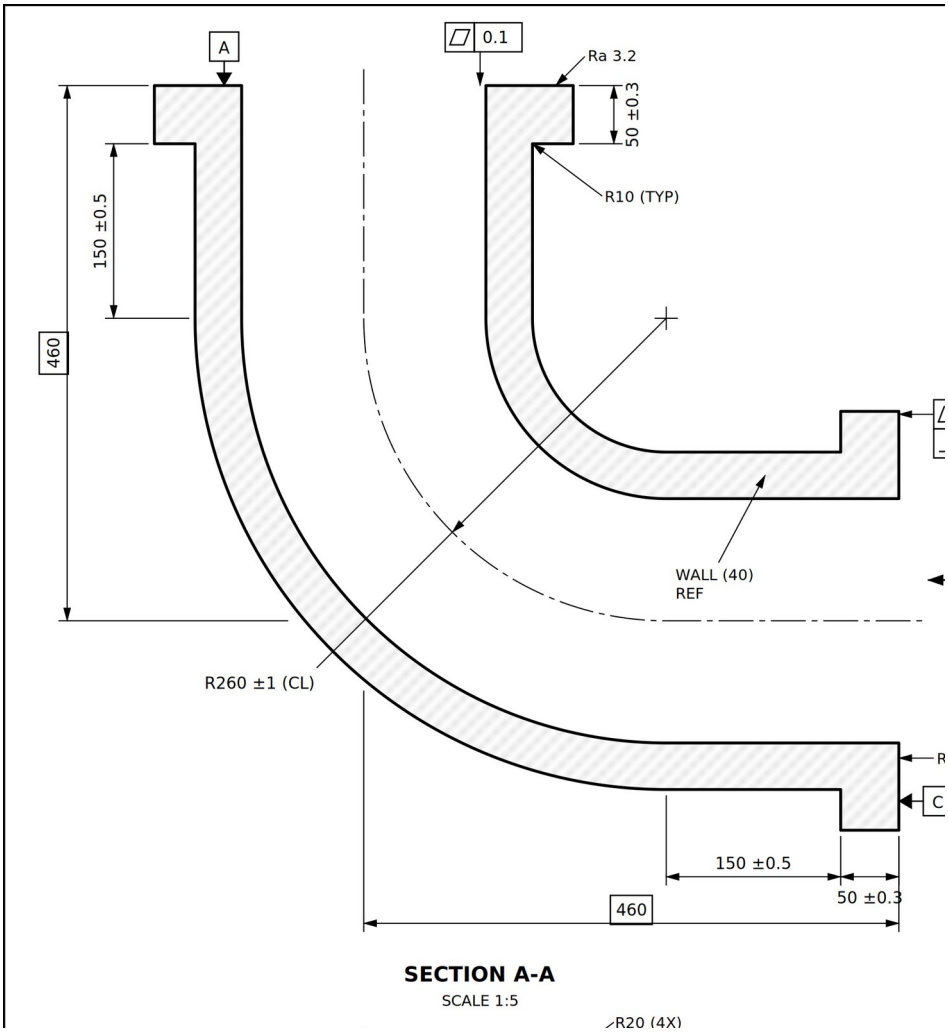


Figure 5. Section A-A: the hatched 40 mm wall, flange thickness 50 ±0.3, straight length 150 ±0.5, bend R260 ±1, basic dimension 460, datums A and C, flatness 0.1 and perpendicularity 0.2.

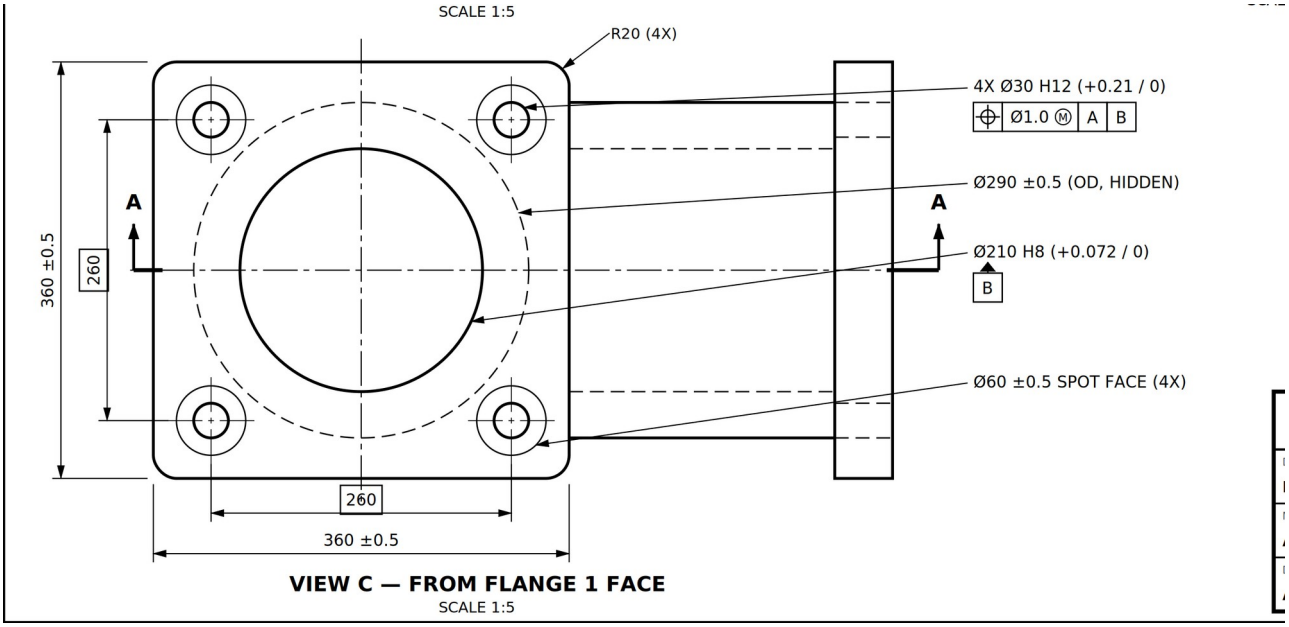


Figure 6. View C, looking at the flange 1 face: 4X Ø30 H12 holes with position Ø1.0 (M) to A | B, Ø60 spot faces, Ø210 H8 bore (datum B), hidden Ø290 outline, basic 260 hole spacing and 360 ±0.5 flange width.

TOLERANCE & GD&T SUMMARY

DATUM A	Flange 1 mating face
DATUM B	Ø210 bore axis, flange 1
DATUM C	Flange 2 mating face
DATUM D	Ø210 bore axis, flange 2
Flange faces (A, C)	Flatness 0.1
Flange 2 face (C)	Perpendicularity 0.2 to A
Bores Ø210 (B, D)	H8 (+0.072 / 0)
4X Ø30 holes per flange	H12 (+0.21 / 0)
Position, flange 1	Ø1.0 (M) | A | B
Position, flange 2	Ø1.0 (M) | C | D
Ø60 spot face	±0.5
Flange thickness 50	±0.3
Flange width 360	±0.5
Straight length 150	±0.5
Bend radius R260	±1.0 (centreline)
Pipe OD Ø290	±0.5
Wall thickness 40	Reference only
Boxed dims (260, 460)	Basic dimensions
Surface finish	Ra 3.2 flange faces
	Ra 12.5 elsewhere
General tolerance	ISO 2768-mK

NOTES

1. All dimensions in mm.
2. Boxed dimensions are BASIC.
3. Unspecified tolerances: ISO 2768-mK.
4. Break sharp edges 0.5 max.
5. Corner radii R20, fillets R10 (as modelled).
6. Holes and spot faces are not cut by A-A.
7. Bolt assumed M27; Ø30 = medium clearance.
8. Floating fastener: T = H − F = 30 − 27 = 3.0 max; Ø1.0 chosen for alignment.
9. Material: AISI 1020 plain carbon steel.

REFERENCE DATA (NOT FOR INSPECTION)

Bolt circle Ø367.7 (REF)
Hole centre to bore axis 183.8 (REF)
Edge distance 50 (REF)
Centreline arc length 408.4 (REF)
Min. wall at 1 MPa ≈ 0.9 mm (design check)

Figure 7. Tolerance and GD&T summary, notes and reference data from the drawing.

8.3 Drawing best-practice checklist

Table 17. Drawing checklist

Practice	How the drawing meets it
Standard sheet and scale	A3 landscape, scale 1:5, units stated as mm
Complete title block	Title, drawing number, material, mass, finish, tolerance, drawn by, date
Datums from function	A to D chosen from the mating faces and bores (Section 7.1)
Basic dimensions boxed	260 hole spacing and 460 overall dimension
No redundant or unclear dimensions	Wall shown as a reference value only; reference data kept in a separate block

Practice	How the drawing meets it
Section view where the inside matters	Section A-A shows the bore and the 40 mm wall
General tolerances stated	ISO 2768-mK note and title block entry
Surface finish stated	Ra 3.2 on flange faces, Ra 12.5 elsewhere

9. Week 1 Feedback and How This Report Addresses It

The Week 1 review asked for more specific data, worked examples and original analysis, with more depth on GD&T, DFM and material selection. The table shows where each point is answered.

Table 18. Feedback tracker

Week 1 feedback	What this report does	Where
Specific data and concrete numbers	Dimensions, tolerances, material properties and a mass are stated throughout	Sections 2 to 7
Worked examples	Mass estimate, wall thickness, Lamé stress, floating fastener, MMC bonus, bore limits	4.3, 5.2, 7.4
Original analysis	The 40 mm wall is 44 times what pressure needs; a 10 mm wall saves about 137 kg and improves bolt access	5.2, 5.3
GD&T principles	Datum scheme with reasons, callouts, size tolerances and inspection plan	Section 7
DFM	Observation, reason and action for each issue; casting and fabrication compared	Section 6
Material selection	Four materials compared and scored; AISI 1020 chosen with its weakness stated	Section 4

10. Challenges, Lessons and Next Steps

Challenges found in the design review

- **Over-built wall.** The 40 mm wall carries about 3 MPa of hoop stress at 1 MPa pressure, so most of the 247 kg is unnecessary.
- **Tight bolt access.** The Ø60 spot face lies only 8.85 mm from the pipe OD.
- **Cost against function.** The tolerances had to be tight enough for alignment (Ø1.0 position, H8 bore) but not so tight that they add machining cost (H12 holes drilled, not reamed).
- **Incomplete first drawing.** It had no tolerances, no datums and an empty title block, so it could not be manufactured from.

Lessons

- Choose datums from how the part works, then add tolerances to features that need them.
- Support each statement with a number; it makes the design checkable.
- Check manufacturability while the design is still easy to change.

Next steps

- Verify the hand-calculated mass against SolidWorks Mass Properties and run DFMXpress on the model.

- Validate the 10 mm wall proposal by finite element analysis under pressure and bolt preload.
- If the part is cast, add draft angles and a machining allowance and prepare a casting drawing.

11. Conclusion

The exercise took a flanged pipe elbow from a dimensioned model to a documented, manufacturable design. AISI 1020 was selected with a weighted comparison, the mass was estimated at about 247 kg, and a datum-based GD&T scheme was applied with a worked assembly margin of 2.0 mm. The design review found that the 40 mm wall is far thicker than pressure requires; a 10 mm wall would save about 137 kg and improve tool access. The final A3 drawing and this report document each step and the reason behind it.

References and standards

- ISO 286-1: limits and fits (IT grades, H8 and H12 values).
- ISO 2768-1 and ISO 2768-2: general tolerances (class m for linear sizes, class K for geometry).
- ISO 273: clearance holes for bolts (medium series for M27).
- ASME Y14.5: GD&T conventions, including the floating-fastener formula and the MMC modifier.
- Typical published material data for AISI 1020, grey cast iron, stainless 304 and aluminium 6061-T6; confirm against the material certificate before manufacture.